MILITARY SPECIFICATION

BEARINGS, PLAIN, SELF-ALIGNING (BeCu, CRES RACE)
GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-B-81936 dated 28 September 1973, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title line: after (BeCu, CRES Race) add "GENERAL SPECIFICATION FOR".

* 2.1 SPECIFICATIONS, Military: Delete "MIL-D-1000 Drawings, Engineering and Associated Lists".

PAGE 2

* 2.1 STANDARDS, Federal: Delete "FED. TEST METHOD, STD. No. 151 Metals; Test Methods" and substitute "FED-STD-151 Metals; Test Methods".

* STANDARDS, Military: Add "DOD-D-1000 Drawings, Engineering and Associated Lists".

Add "MIL-STD-1599 Bearings, Control System Components and Associated Hardware used in the Design and Construction of Aerospace Subsystems".

PAGE 3

* After 3.1.1, insert the following new paragraph:

"3.1.2 Product Manufacture. Except for the swaging operation, the manufacturer is permitted to sub-contract manufacturing operations without violating the requirements of 3.1.1. The swaging operation shall be performed in the plant listed on the Qualified Products List."

3.4.2, line 2: Delete and substitute "rating (RHR) in accordance with ANSI B46.1 and not in excess of the values shown on the applicable".

FSC 3120

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* After 3.4.4 add the following new paragraph:

"3.4.5 Conformity. The conformity between the ball and race shall be checked in accordance with 4.6.6. Measurements shall be taken at a minimum of six positions, three uniformily spaced positions on each side of the lubrication groove. Ten percent of the race width on each edge shall be excluded from measurement and in addition, for sizes -12 and greater, 5% of the race width on each side of the lubrication groove shall be excluded. Variation between measured values shall not exceed 0.001 inch. Overforming within 10% of each side face (H/10) is not permitted. Underforming is not controlled."

PAGE 3

3.5.1, line 1: Delete title "Radial static load" and substitue "Radial static limit load".

3.5.2, line 1: Delete title "Axial static load" and substitute "Axial static limit load".

* 3.6, line 4: Delete "MIL-D-1000" and substitute "DOD-D-1000".

PAGE 5

3.8, line 2: Delete "griding" and substitute "grinding".

* Delete Table II and substitute the following:

"TABLE II QUALIFICATION TESTS

Examinations and Tests	Paragraph No.	Samples to be Tested (1)
Examination of Product	4.5.1	5
Radial Static Limit Load	4.6.1	3
Axial Static Limit Load	4.6.2	3
Mode I Dynamic Test	4.6.3.1	3
Mode II Dynamic Test	4.6.3.2	3 (2)
Conformity	4.6.6	3

(1) Sizes M81936/1-6, -12 and -20 only.

(2) At manufacturer's option, the Mode II tests may be performed with M81936/1-6R, -12R and -20R bearings."

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* 4.3.1, line 2: Delete" fifteen (15) bearings" and substitute "eighteen (18) bearings".

PAGE 6

* 4.3.4, delete the contents of paragraph 4.3.4, and substitute:

"4.3.4 Retention of qualification. The continued listing of a product on the Qualified Products List is dependent upon a periodic verification of the manufacturer's continued compliance with the requirements of this specification and with standardization regulations. As part of that verification process, each manufacturer must complete DD Form 1718 during October of each odd numbered year. This form, supplied by the qualifying activity, is to be signed by a responsible official of management and sent to the Naval Air Engineering Center, SESD Code 9311, Lakehurst, NJ 08733-5100."

* 4.6.5, line 2: Delete "Federal Test Method Standard No. 151." and substitute "FED-STD-151."

* After Para 4.6.5 add the following new paragraph:

"4.6.6 Conformity. Encapsulate the bearing in plastic material, as used in metallurgical mounts, to prevent motion to the ball with respect to the race. Section the bearing on a diameter and normal to the race side face to produce a surfce as in figure 4(A) or 4(B). Grind and polish the face to obtain a true view of the edges of the curved portions of the ball and race. By use of a microscope with XY table or other suitably accurate technique, measure dimension "t" radially from the ball to the outer race in a minimum of six positions (three uniformly spaced positions on each side of the lubrication groove). The measurements need not be taken closer to the race face than 10% of "H". In addition, for bearing sizes -12 and greater, readings need not be taken in the area within 5% of the race width on each side of the lubrication groove. Overforming of the outer race is defined as the condition when the side portions of the outer race are in closer proximity to the ball surface than the middle portion of the outer race. Overforming within 10% of each side face (H/10) is not permitted. Underforming is not controlled."

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PAGE 11

* After Figure 3 add Figures 4(A) and 4(B)

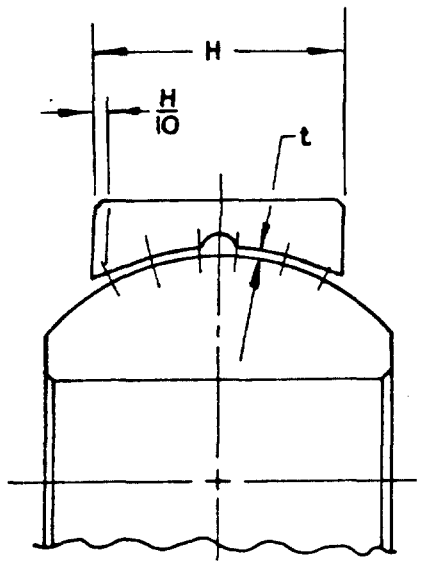


FIGURE 4(a).
Conformity sizes
thru -10

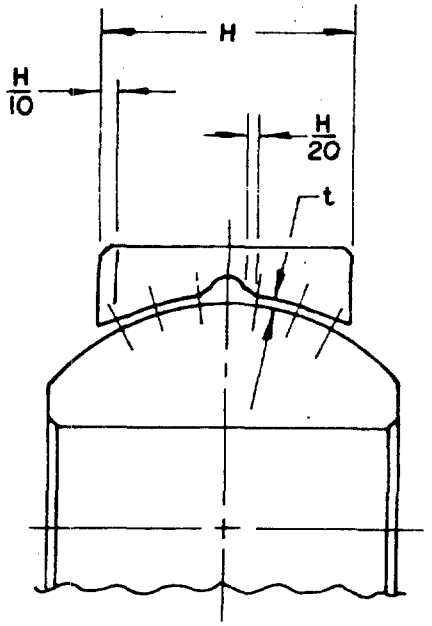


FIGURE 4(b)
Conformity sizes
-12 and greater

Note (1) Overforming in the H/10 region is not allowed. Underforming is not controlled.

Note (2) For sizes -3 through -10, location of the two conformity measurements adjacent to the grease groove shall be within 0.0050 inch of the groove edge.

PAGE 12

5.2, line 4: Delete "Antifriction, Type (I, II, III, IV)".

6.1, line 4: Add "For specific design information on the capability of these bearings under particular load, speed and motion conditions, the user is referred to MIL-STD-1599."

6.3, line 12: Delete "30211" and substitute "6061".

Delete line 13.

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